LECANEMAB FOR EARLY ALZHEIMER'S DISEASE: 48-MONTH RESULTS FOR APOE ε4 NON-CARRIERS OR HETEROZYGOTES FROM THE CLARITY AD OPEN-LABEL EXTENSION

Lutz Froelich,[1] Shobha Dhadda,[2] Michio Kanekiyo,[2] Amanda Goodwin,[2] Mark Hodgkinson,[3] Steven Hersch,[2] Michael Irizarry,[2] and Lynn D. Kramer[2]

1. Medical Faculty Mannheim, University of Heidelberg, Central Institute of Mental Health, Mannheim, Germany. 2. Eisai Inc. Nutley, NJ, USA. 3. Eisai Europe Ltd - Hatfield (United Kingdom).

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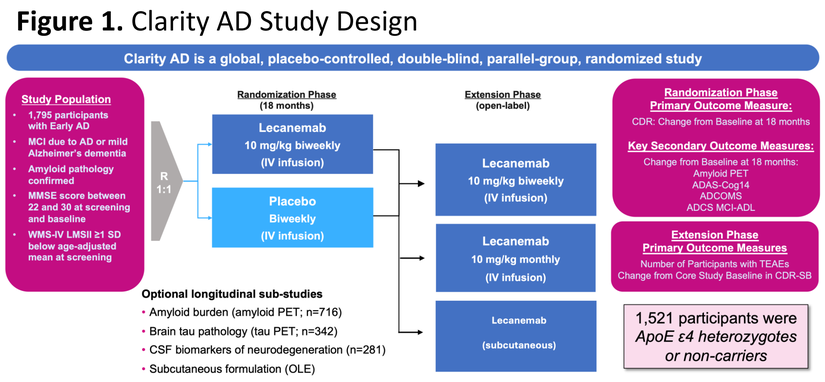
Introduction

- Approximately 7 million people in Europe are currently living with Alzheimer's disease (AD), with estimates predicting a rise to 14 million by 2030[1-2]
- Lecanemab is a humanized immunoglobulin G1 monoclonal antibody that selectively binds to soluble Aβ aggregate species[3]
- In Clarity AD, lecanemab 10 mg/kg biweekly treatment reduced markers of amyloid in participants with early AD and resulted in less decline on measures of cognition and function versus placebo consistent with disease modification[4]
- Lecanemab was shown to be well tolerated in multiple clinical trials, although an increased rate of amyloid-related imaging abnormalities (ARIA) was observed, especially in individuals who were ApoE ε4 homozygotes[4-7]
- Here, we present the long-term results from Clarity AD participants who were ApoE ε4 non-carriers or heterozygotes

Methods

- Clarity AD is an 18-month, phase 3, multicenter, double-blind trial in subjects with early AD (mild cognitive impairment or mild AD) and confirmed amyloid positivity, with an open-label extension phase (OLE) phase (**Figure 1**)[4]

Figure 1. Clarity AD Study Design



Amyloid imaging related abnormalities (ARIA) occurrence was monitored throughout the study by central reading of magnetic resonance imaging.

- Following the 18-month, randomized core phase of Clarity AD, eligible participants received open-label lecanemab in the OLE
- Cognitive and functional efficacy assessments included Clinical Dementia Rating – Sum of Boxes (CDR-SB), Alzheimer's Disease Assessment Scale-Cognitive Subscale 14 (ADAS-Cog14) and Alzheimer's Disease Cooperative Study-Activities of Daily Living Scale for Mild Cognitive Impairment (ADCS MCI-ADL)
- Time saved and time to progression to next stage of AD with lecanemab treatment was also evaluated through 48 months
- CDR-SB efficacy results were compared to a matched Alzheimer's Disease Neuroimaging Initiative (ADNI) cohort obtained from the ADNI database (adni.loni.usc.edu) as well as a historical control composed with data from the Swedish BioFINDER study (NCT01208675). Control populations were selected based on key eligibility criteria for Clarity AD

Results

- Overall, 1,795 patients with early AD (MCI or mild dementia due to AD, with confirmed presence of amyloid pathology) were enrolled in Clarity AD, of which 1,521 were ApoE ε4 non-carriers or heterozygotes (**Table 1**)
- Of the total number of patients randomized 31% were non-carriers, 53% were heterozygotes and 16% were homozygotes

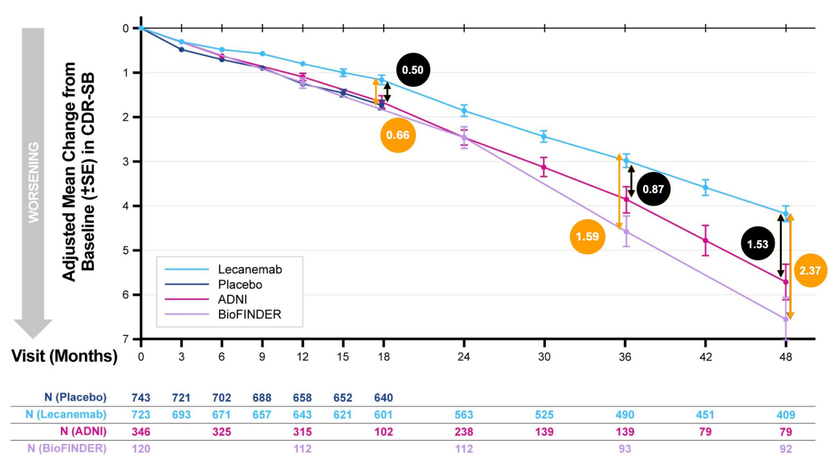
Table 1. Baseline Characteristics

	Clarity AD	ADNI	BioFINDER
N	1521	346	120
Age			
Mean (SD)	71.8 (7.95)	74.6 (7.33)	73.1 (4.74)
Median	73.0	75.3	73.9
APOE4 Carrier Status			
Positive n (%)	957 (62.9)	215 (62.1)	75 (62.5)
Negative n (%)	564 (37.1)	131 (37.9)	45 (37.5)
Disease Stage[a], n (%)			
MCI due to AD	938 (61.7)	222 (64.2)	120 (100)
Mild AD	583 (38.3)	124 (35.8)	0
Baseline CDR-SB			
Mean (SD)	3.20 (1.344)	2.61 (1.628)	1.46 (0.81)
Median	3.00	2.00	1.50

Results *(continued)*

- CDR-SB results in ApoE ε4 non-carriers or heterozygotes of Clarity AD are shown in **Figure 2**

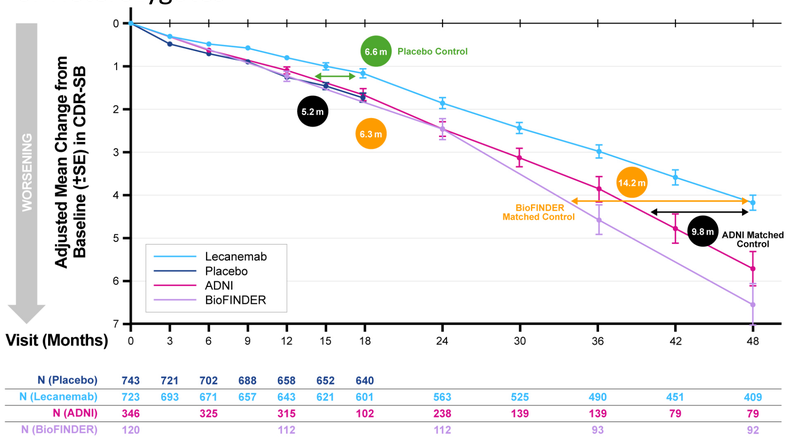
Figure 2. CDR-SB in ApoE ε4 Non-Carriers or Heterozygotes



Note: OLE includes those participants on intravenous and subcutaneous formulations. BioFINDER data are from BioFINDER 1. Analysis done in ApoE ε4 Non-carriers or Heterozygotes. In the EU, lecanemab is indicated as an intravenous infusion at a dosage of 10 mg/kg body weight, administered once every 2 weeks.

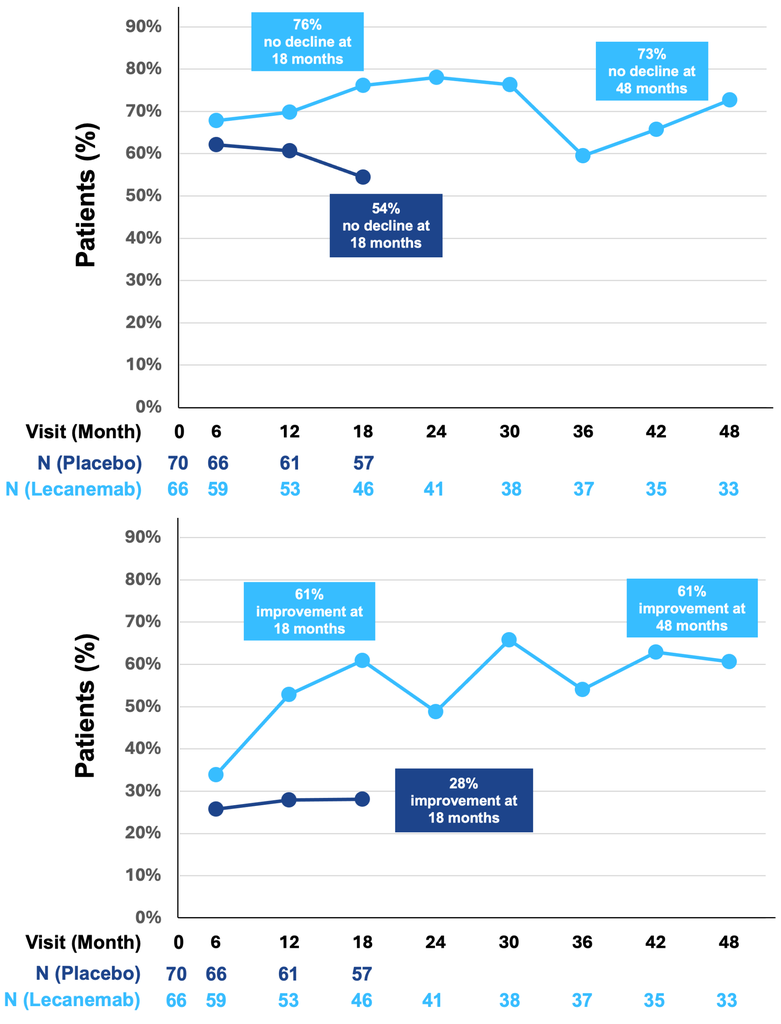
- Lecanemab-treated participants continued to accrue benefit in CDR-SB through 48 months (**Figure 2**)
- Secondary endpoint results for ADAS-Cog14 and ADCS MCI-ADL were consistent with the primary endpoint
- Time saved evaluated at 48 months for CDR-SB between lecanemab was 9.8 and 14.2 months vs ADNI and BioFINDER control, respectively (**Figure 3**)

Figure 3. CDR-SB Time Saved Analysis in ApoE ε4 Non-Carriers or Heterozygotes



- The subgroup of subjects with no/low tau & low amyloid at baseline continued to benefit through 48 months (**Figure 4**)

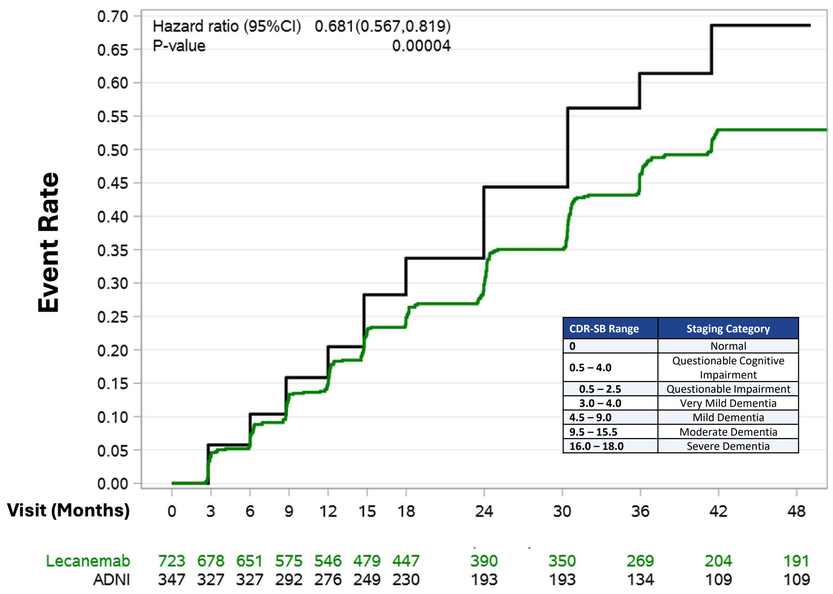
Figure 4. CDR-SB Response Rates (No Decline and Improvement) in Participants with No/Low Tau PET



Efficacy was also assessed in the subgroup of participants with lower pathology—considered to represent an early pathologic stage of disease—as defined as having either low or no tau (Cutoffs for MK Tau PET: No/Low SUVR <1.06; Intermediate SUVR 1.06-2.91; High SUVR >2.91).

- Lecanemab reduced the risk of progression to next AD stage by 32% compared to the ADNI natural history cohort (**Figure 5**)
- Results for delaying the next stage of dementia (ie, moderate or severe AD) also favored lecanemab, with a relative risk of progression delayed with lecanemab treatment by 55% vs ADNI

Figure 5. Time to Worsening on CDR-SB



Progression to next stage of dementia was defined as CDR-SB Score progressing from MCI (0.5-4) to mild AD dementia (4.5-9) or moderate dementia (9.5-15.5) based on dementia staging on CDR-SB (O'Bryant et al., Arch Neurol 2008). Progression to dementia was defined as CDR-SB Score progressing from mild AD dementia (4.5-9) to moderate dementia (9.5-15.5) or severe dementia (16.0-18.0) based on dementia staging on CDR-SB (O'Bryant et al., Arch Neurol 2008). Given less frequent assessment, since controlled-based imputation was used for missing data in this analysis, CDR-SB (which has greater range) was used rather than global CDR for disease staging

- No new safety signals were observed across exposure in the Clarity AD Core+OLE of 3478 person-years (mean 2.5 years; >280 with 4 years (**Table 2**)

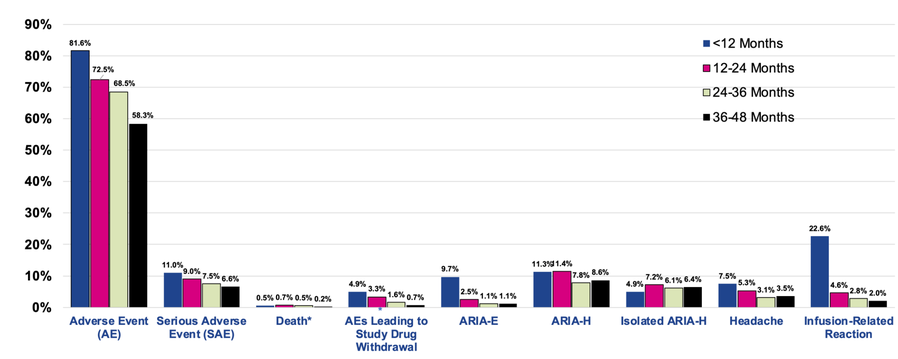
Table 2. Summary of Adverse Events and ARIA (Exposure Adjusted)

	Placebo N = 764			Lecanemab N = 757			Lecanemab (Double-Blind + OLE) N = 1366		
	n	%	Exposure Adjusted*	n	%	Exposure Adjusted*	n	%	Exposure Adjusted*
Adverse Event (AE)	620	**81.2**	59.3	667	**88.1**	66.9	1250	**91.5**	35.9
Serious Adverse Event (SAE)	86	**11.3**	8.2	112	**14.8**	11.2	309	**22.6**	8.9
Death[a]	8	**1.0**	0.8	7	**0.9**	0.7	29	**2.1**	0.8
Deaths with concurrent ARIA or ICH irrespective of ARIA being cause of death	1	**0.1**	<0.1	0	**0**	0	1	**0.1**	<0.1
AEs Leading to Study Drug Withdrawal	23	**3.0**	2.2	45	**5.9**	4.5	126	**9.2**	3.6
ARIA-E	10	**1.3**	1.0	67	**8.9**	6.7	151	**11.1**	4.3
ARIA-H	52	**6.8**	5.0	98	**12.9**	9.8	292	**21.4**	8.4
Isolated ARIA-H	45	**5.9**	4.3	61	**8.1**	6.1	187	**13.7**	5.4
ICH[a]	2	**0.3**	0.2	4	**0.5**	0.4	8	**0.6**	0.2

In the EU Lecanemab is indicated as an intravenous infusion at a dosage of 10 mg/kg body weight, administered once every 2 weeks. ***Exposure adjusted rate (per 100 persons per year)**
n, %, exposure-adjusted rate (per subject-year) are presented. OLE is based on IV datasets (as of 31 Mar 2025).
Analysis done in ApoE ε4 Non-carriers or Heterozygotes.
a: Includes all post-treatment events. ARIA-E, amyloid related imaging abnormalities - edema; ARIA-H, ARIA with hemosiderin deposits; ICH, intracerebral hemorrhage.

- Adverse event rates were lower or stabilized each additional year (**Figure 6**)

Figure 6. Summary of Adverse Events by 12-Month Interval



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Conclusions

- In the ApoE ε4 non-carriers or heterozygotes population of Clarity AD, the long-term safety profile of lecanemab was confirmed, with no new safety signals
- Ongoing lecanemab in participants treated through 48 months was associated with increasing treatment difference versus matched placebo data, consistent with a disease-modifying effect

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